

Maincrafts — Business Performance Report

Data Analytics & Business Intelligence — Task 2

This report presents a structured business analysis using two related datasets: Orders and Customers. The workflow follows the task requirements: SQL JOINS and aggregations, KPI calculations, comparative analysis, Excel-style analysis, and a KPI dashboard. The dataset included with this submission is synthetic and was created to demonstrate the complete workflow.

KPI	Result
Total Sales	\$1,030,154.49
Total Profit	\$137,981.18
Profit Margin	13.39%
Total Customers	120
Total Orders	900
Average Order Value	\$1,144.62

1. Regional Performance

The North region generates the highest revenue at \$317,453.01. Its profit margin is 13.29%. Regional performance should be compared using both sales volume and margin so that high-revenue but low-profit regions are not over-prioritized.

Region	Sales	Profit	Margin
North	\$317,453	\$42,181	13.3%
West	\$270,429	\$36,592	13.5%
South	\$259,270	\$34,571	13.3%
East	\$183,003	\$24,637	13.5%

2. Category Performance

Office Supplies has the highest profit margin at 19.20%. Category-level margin is calculated as total profit divided by total sales, matching the KPI definition requested in the task. Management should protect high-margin categories while reviewing low-margin categories for discounting and cost issues.

Category	Sales	Profit	Margin
Office Supplies	\$144,457	\$27,733	19.2%
Technology	\$506,849	\$78,283	15.4%
Accessories	\$88,815	\$12,338	13.9%
Furniture	\$290,034	\$19,626	6.8%

3. Customer & Segment Analysis

The Consumer segment contributes the highest sales among customer segments at \$535,978.35. The Top 10 customer table in the Excel workbook and dashboard can be used to identify accounts that contribute disproportionately to revenue. These customers are suitable targets for retention, cross-selling, and account-management initiatives.

Segment	Sales	Profit	Margin
Consumer	\$535,978	\$71,584	13.4%
Corporate	\$281,696	\$36,539	13.0%
Home Office	\$212,480	\$29,858	14.1%

4. Discount & Profitability

The lowest observed profit margin occurs at a 25% discount level (7.52%), while the strongest margin in the discount comparison is at 0% (20.95%). The relationship should be monitored before introducing broad discount campaigns; discounting can increase sales volume while reducing the quality of revenue.

5. Sales Trend & Seasonality

The monthly trend shows recurring month-to-month variation rather than a perfectly flat sales pattern. The average monthly sales in the second half of the period are 2.7% relative to the first half. This indicates that management should use monthly trend monitoring to identify recurring peaks and troughs and align inventory, promotions, and staffing accordingly.

6. Recommendations for Business Growth

- Prioritize the highest-revenue region while diagnosing why other regions lag in sales or margin.
- Protect high-margin categories and review discounting, pricing, and cost structure for lower-margin categories.
- Create retention and cross-sell programs for the highest-value customers.
- Use monthly KPI monitoring to detect seasonal peaks and prepare inventory and marketing campaigns earlier.
- Track Profit Margin alongside Total Sales so growth is evaluated for quality, not only volume.
- Use the dashboard filters in a BI implementation to compare Region, Product Category, Date, and Customer Segment.

Submission note: The included dataset is synthetic, so the numerical findings are demonstration results for the completed Task 2 workflow.